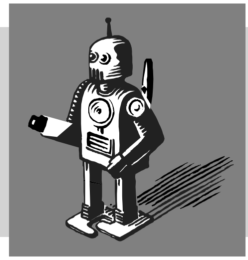
3

Actuators and Grippers

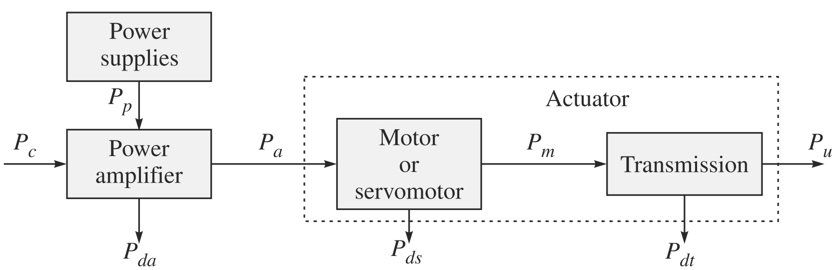


Actuators are devices which drive a robot including its grippers. They are like the muscles of a human arm and hand. While the human arm provides motion, the hand is used for object manipulation. As illustrated in Fig. 2.3, the same is true with a robot. Its arm, be Cartesian or anthropomorphic or of other type, which are explained in Section 2.2, provides motion, whereas its gripper, which is equivalent to a human hand, manipulates objects. Both require some kind of actuator system powered by electricity or compressed air or pressurized fluid. These systems are presented in this chapter, along with different types of grippers used in robotic applications.

An actuator system comprises of several subsystems, namely,

(i) a power supply;
(ii) a power amplifier;
(iii) a servomotor; and
(iv) a transmission system.

The connections between all the actuator components are depicted in Fig. 3.1.



P_p: Primary source of power (electricity or pressurized fluid or compressed air); P_c: Input control power (usually electric); P_a: Input power to motor (electric or hydraulic or pneumatic type); P_m: Power output from motor; P_u: Mechanical power required; P_{da}, P_{ds}, and P_{dt}: Powers lost in dissipation for the conversions performed by the amplifier, motor, and transmission

Fig. 3.1 An actuator system

To choose an actuator, it is worth starting from the requirements imposed on the mechanical power P_u by the force and velocity that describe the joint motion. Based on the source of input power P_a, the actuators can be classified into three groups:

> **Actuator vs. Motor**
>
> A motor together with transmissions and other accessories, if any, is referred as an actuator (see Fig. 3.1). However, in many instances, they are used interchangeably to mean that they move a robot link.

1. Electric Actuators The primary input power supply is the electric energy from the electric distribution system.

2. Hydraulic Actuators They transform hydraulic energy stored in a reservoir into mechanical energy by means of suitable pumps.

3. Pneumatic Actuators They utilize pneumatic energy, i.e., compressed air, provided by a compressor and transform it into mechanical energy by means of pistons or turbines.

A portion of the motor input power P_a is converted to the output as a mechanical power P_m, and the rest P_{ds} is dissipated because of mechanical, electrical, hydraulic, or pneumatic motor losses. In a robotic application, an actuator should have the following characteristics:

- Low inertia
- High power-to-weight ratio
- Possibility of overload and delivery of impulse torques
- Capacity to develop high accelerations
- Wide velocity ranges
- High positioning accuracy
- Good trajectory tracking and positioning accuracy

Based on wide application of electric actuators in industrial robots and their easy availability in the stores next door selling toys or repairing photocopiers, washing machines, etc., they are presented first. It will be followed by hydraulic and pneumatic actuators. While hydraulic actuators are suitable for applications where the requirement is high power-to-weight ratio, pneumatic actuators are often used by electrically actuated robots for their grippers requiring only on-off motions of the jaws. Use of a pneumatic gripper makes the robot system a little lighter and cost effective.

Who is ABB?

ABB is a Zurich, Switzerland-based multinational company operating in power and automation technology areas. It was formed in 1988 through the merger of ASEA of Sweden and Brown, Boveri and Cie of Switzerland. It has operations in more than 100 countries, and is one of the major industrial robot manufacturers in the world.

3.1 ELECTRIC ACTUATORS

Electric actuators are generally those where an electric motor drives robot links through some mechanical transmission, e.g., gears, etc. In the early years of industrial robotics, hydraulic robots were the most common, but recent improvements in electric-motor design have meant that most new robots are of all-electric construction. The first commercial electrically driven industrial robot was introduced in 1974 by ABB. The advantages and disadvantages of an electric motor are the following:

Advantages

- Widespread availability of power supply.
- Basic drive element in an electric motor is usually lighter than that for fluid power, i.e., pressurized fluid or compressed air.
- High power-conversion efficiency.

- No pollution of working environment.
- Accuracy and repeatability of electric drive robots are normally better than fluid power robots in relation to cost.
- Being relatively quiet and clean, they are very acceptable environmentally.
- They are easily maintained and repaired.
- Structural components can be lightweight.
- The drive system is well suited to electronic control.

Disadvantages

- Electrically driven robots often require the incorporation of some sort of mechanical transmission system.
- Additional power is required to move the additional masses of the transmission system.
- Unwanted movements due to backlash and plays in the transmission elements.
- Due to the increased complexity with the transmission system, we need complex control requirement and additional cost for their procurement and maintenance.
- Electric motors are not intrinsically safe, mainly, in explosive environments.

The above disadvantages are gradually being overcome with the introduction of direct-drive motor system, in which the electric motor is a part of the relevant robot arm joint, thus, eliminating the transmission elements. Moreover, the introduction of newer brushless motors allow electric robots to be used in some fire-risk applications such as spray painting, as the possibility of sparking at the motor brushes is eliminated. Different types of electric motors are stepper motors, and the dc and ac motors, which are explained next.

dc vs. ac

dc stands for *direct current*, whereas ac means *alternating current*. Batteries in flash lights, cars, etc., supply dc. For domestic use, however, the electric supply is ac with 230 volts, 50 cycles (in India and other countries), 110 volts, 60 cycles (USA and others), and similar.

3.1.1 Stepper Motors

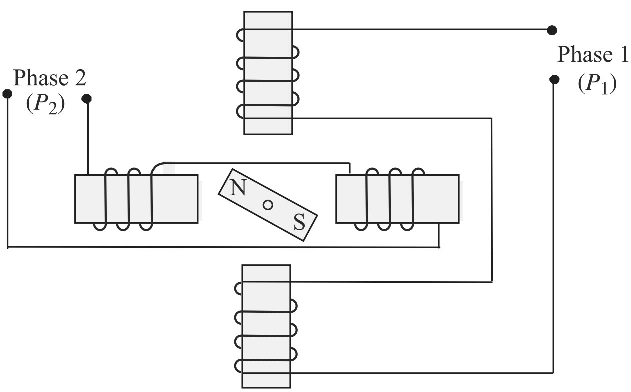
Stepper motors (also called stepping motors or step motors) were first used for remote control of the direction indicators of torpedo tubes and guns in British warships and later, for a similar purpose, in the US Navy. A variable reluctance-type stepper motor was first patented in 1919 by C.L. Walker, a Scottish civil engineer. However, its commercial production did not start until 1950.

Such motors are used with small and medium robots and with teaching and hobby robots. These motors are widely used in other industrial applications mainly due to their advantage of not requiring any feedback system and, accordingly not incurring the associated cost. Stepper motors are, however, compatible with many feedback devices, if so desired, and are used in full servo-control configurations in medium-duty industrial robots. Because they are digitally controlled, they can be referred to as *digital motors* or *actuators*. These motors do not require the expense of digital-to-analog conversion equipment when connected to a computer-control system. A commercially available stepper motor is shown in Fig. 3.2.

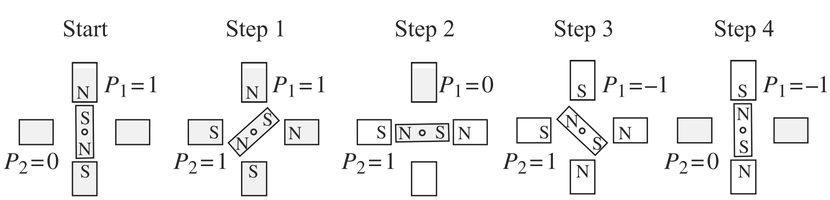
[Courtesy: http://www.polulu.com]

Fig. 3.2 Stepper motor (Bipolar, 200 Steps/Rev, 20 × 30 mm, 3.9 V, 0.6 A/Phase)

Normally, the shaft of a stepper motors rotates incrementally in equal steps in response to a programmed input pulse train. As shown in Fig. 3.3(a), current in any of the two phases, i.e., with P_1 and P_2, will magnetize the pair into north and south poles, indicted with N and S, respectively. Accordingly, the permanent magnet at the centre will rotate in order to align itself along a particular phase, which is demonstrated in Fig. 3.3(b). Switching the currents in the two phases in an appropriate sequence can produce either clockwise (CW) or counterclockwise (CCW) rotations, as tabulated in Table 3.1. The switching sequence of Table 3.1 corresponds to what is known as *half-stepping* with a step angle of 45°, whereas the full-stepping corresponds to 90° in which one phase is energized at a time. See Example 3.3. It is pointed out here that



(a) A two-phase stepper motor



(b) Half-stepping sequence (both phases energized simultaneously)

Fig. 3.3 Working principle of a two-phase stepper motor one can also achieve micro-stepping (non-identical steps) up to 1/125 of full-step by changing the currents in small steps instead of switching them on and off, as in the case of half- and full-stepping. While micro-stepping is advantageous from the point of view of accurate motion control using a stepper motor, it has the disadvantage of reduced motor torque.

Note that the steps are achieved through phase activation or switching sequences triggered by the pulse sequences. The switching logic that decides the states of the phases of a given step can be generated using a look-up table given in Table 3.1. The same sequences can also be generated using a logic circuitry which is typically an electronic device.

Table 3.1 Half-stepping sequence of a two-phase stepper motor (both phases energized simultaneously)

Phases \ Step	1	2	3	4	5	6	7	8	Rotation
	→	→	→	→	→	→	→	→	CW
P_1	1	0	–1	–1	–1	0	1	1	
P_2	1	1	1	0	–1	–1	–1	0	
	←	←	←	←	←	←	←	←	CCW

As the rotor indexes round a specific amount for each control pulse, any error in positioning is noncumulative. To know the final position of the rotor, all that is required is to count the number of pulses fed into the stator's phase winding of the motor. The number of pulses per unit time determines the motor speed. The rotor can be made to index slowly, coming to rest after each increment or it can move rapidly in a continuous motion termed *slewing*. Maximum dynamic torque in a stepper motor occurs at low pulse rates. Therefore, it can easily accelerate a load. Once the required position is achieved and the command pulses cease, the shaft stops without the need for clutches or brakes. The actual rotational movements or step angles of the shaft are obtainable typically from 1.8° to 90° depending on the particular motor choices. Thus, with a nominal step angle of 1.8°, a stream of 1000 pulses will give an angular displacement of 1800° or five complete revolutions. They have also a low velocity capability without the need for gear reduction. For instance, if the previously mentioned motor is driven by 500 pulses per second, it will rotate at 150 rpm. Other advantages of the stepper motor are that the motor inertia is often low, and also if more than one stepper motor is driven from the same source then they will maintain perfect synchronizations.

Some disadvantages of stepper motors are that they have a lower output and efficiency compared to other motors, namely, dc and ac, and drive inputs and circuitry have to be carefully designed in relation to the torque and speed required. There are various types of stepper motors which are based on the nature of a motor's rotor, e.g., variable reluctance, permanent magnet, and hybrid. A good account of stepper motors is given in de Silva (2004), whereas an online video showing their working principles are available in WR-Stepper (2013)[1]. A typical specification of a stepper motor is given in Table 3.2, whereas a general torque-speed plot is shown in Fig. 3.4.

Table 3.2 Specifications of a stepper motor [Courtesy: www.robokits.co.in]

Brand and Model	NEMA 23, RMCS-115
Step angle	1.8°
Configuration	6 wire stepper motor
Holding torque	10 kg-cm bipolar
Rated voltage	3.3 VDC
Phase current	2 Amp
Resistance/Phase	1.65 E
Inductance/Phase	2.2 mH
Rotor inertia	275 gcm^2
Detent torque	0.36 kg-cm
Length	51 mm
Weight	650 grams

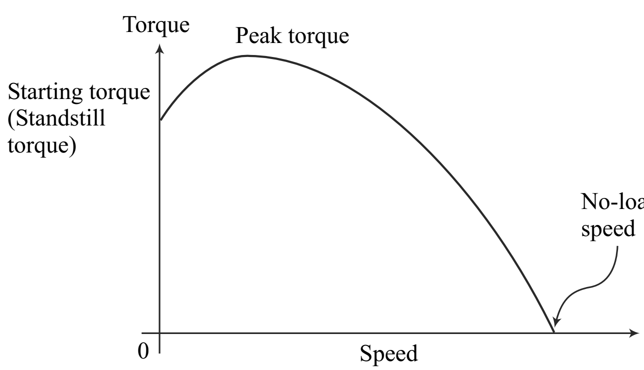


Fig. 3.4 Torque-speed characteristics of a stepper motor

1. Variable-Reluctance Stepper Motor *Magnetic reluctance,* or simply *reluctance,* is the analog of electrical resistance. Just as current occurs only in a closed loop, so magnetic flux occurs only around a closed path, although this path may be more varied than that of the current. Figure 3.5(a) shows the basic form of the variable-reluctance stepper motor. The rotor is made of soft steel and it has four poles, whereas the stator has six

Detent Position

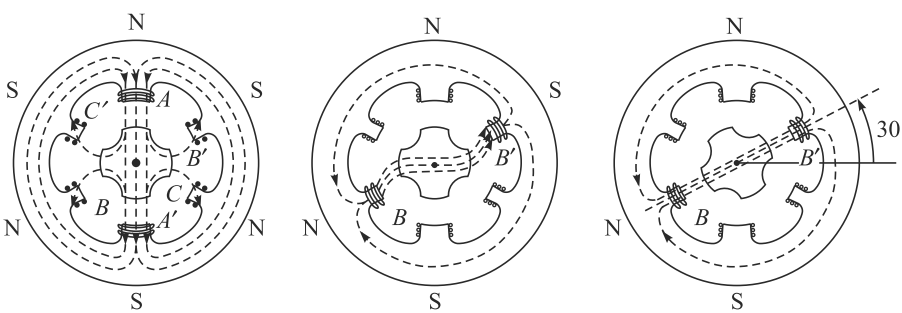
It is the equilibrium configuration when the rotor assumes the minimum reluctance position for that step.

1 WR stands for Web Reference given at the end of the list of References. They mainly contain the links to online materials including videos, which are expected to enhance the understanding of a concept to the reader.

poles. When one of the phases, say AA', is excited due to a dc current passing through the coils around the poles, the rotor positions itself to complete the flux path shown in Fig. 3.5(a). Note that there is a main flux path through the aligned rotor and stator teeth, with secondary flux paths occurring as indicated. When rotor and stator teeth are aligned in this manner, the reluctance is minimized and the rotor is at rest in this position. This flux path can be considered rather like an elastic thread and always trying to shorten itself. The rotor will move until the rotor and stator poles are lined up. This is termed as the position of minimum reluctance. To rotate the motor counterclockwise, the phase AA' is turned off and phase BB' is excited. At that point, the main flux path has the form indicated in Figs. 3.5(b-c). This form of a stepper motor generally gives step angles of 7.5° or 15°, which are referred as *half-stepping* and *full-stepping*, respectively. Note that a disadvantage of variable-reluctance stepper motors is that it has zero holding torque when the stator windings are not energized (power off) because the rotor is not magnetized. Hence, it has no capacity to hold a load in power-off mode unless mechanical brakes are employed.

Holding Torque vs. Detent Torque

Holding torque is the amount of torque that the motor produces when it has rated current flowing through the windings but the motor is at rest, whereas the *detent torque* is the amount of torque that the motor produces when it is not energized, i.e., no current is flowing through the windings.



(a) Basic configuration (b) Beginning of step (c) Completed step

Fig. 3.5 Variable-reluctance stepper motor

2. Permanent-Magnet Stepper Motor The basic method of operation of a permanent-magnet type is similar to the variable-reluctance type. As illustrated in Fig. 3.6, there are two coils A and B, each of them having four poles but displaced from each other by half a pole pitch. The rotor is of permanent-magnet construction and has four poles. Each pole is wound with field winding, the coils on opposite pairs of poles being in series. Current is supplied from a dc source to the winding through switches. It can be seen in Fig. 3.6(a) that the motor is at rest with the poles of the permanent magnet rotor held between the

Harmonic drive and Ratchet-and-pawl vs. Stepper motors

Harmonic drive and *ratchet-and-pawl* are two transmission mechanisms which produce intermittent motions purely through mechanical means. They can be loosely categorized as stepper motors.

residual poles of the stator. In this position, the rotor is locked unless a turning force is applied. If the coils are energized and, in the first pulse, the magnetic polarity of the poles of the coil *A* is reversed, the rotor will experience a torque and will rotate counterclockwise, as shown in Fig. 3.6(b). The reverse poles are shown as *A′*.

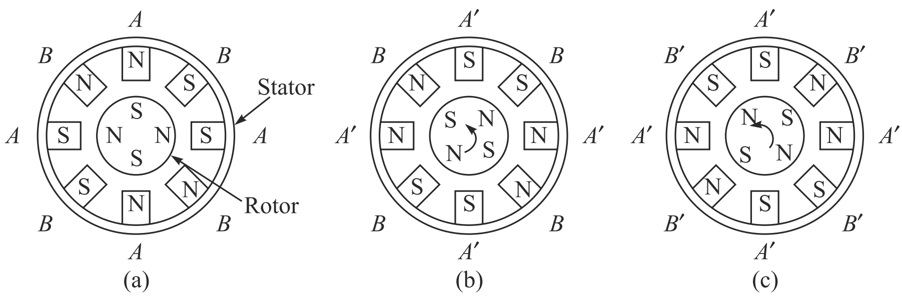


Fig. 3.6 Permanent-magnet stepper motor

If the coil *B* poles are now reversed to *B′*, as shown in Fig. 3.6(c), the rotor will again experience a torque, and once more the poles of the rotor are positioned midway between the stator poles. Thus, by switching the currents through the coils, the rotor rotates by 45°. If in the first pulse, the poles of the coil *B* had been reversed then the motor would have rotated clockwise. With this type of motor, commonly produced step angles are 1.8°, 7.5°, 15°, 30°, 34°, 90°.

3. Hybrid Stepper Motor Hybrid stepper motors are the most common variety of stepper motors in engineering applications. They combine the features of both the variable reluctance and permanent-magnet motors, having a permanent magnet encaged in iron caps which are cut to have teeth, as shown in Fig. 3.7. A hybrid stepper motor has two stacks of rotor teeth on its shaft (WR-Stepper, 2013). The two rotor stacks are magnetized to have opposite polarities, while two stator segments surround the two rotor stacks. The rotor sets itself in the minimum reluctance position in response to a pair of stator coils being energized. Typical step angles are 0.9° and 1.8°.

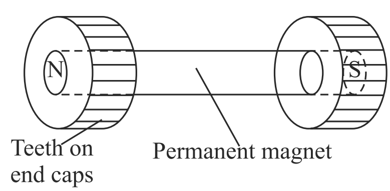


Fig. 3.7 Hybrid stepper motor

From the descriptions of stepper motors, it is, therefore, apparent that the rate at which the pulses are applied determines the motor speed, the total number of pulses determines the angular displacement, and the order of energizing the coils in the first instance determines the direction of rotation. It is because of this ease of driving using direct digital control that stepper motors are well suited for use in a computer-controlled robot, although the motor does require interfacing with a high-current pulse source.

Example 3.1 Pitch Angles of Rotors and Stators

If the angles of two adjacent teeth of a rotor and a stator, called the pitch angles, are denoted by θ_r and θ_s, respectively, they can be calculated as

$$\theta_r = \frac{360°}{n_r} \text{ and } \theta_s = \frac{360°}{n_s} \qquad (3.1)$$

where by n_r and n_s are the corresponding number of teeth in rotors and stators.

Stepper Motor Model

Model of a stepper motor is described by its torque expression with respect to its rotor angle, i.e., $\tau = -\tau_{max} \sin(n_r\theta)$, where τ_{max} is maximum torque during a step (holding torque), n_r is the number of rotor teeth, and θ is any angular position.

Example 3.2 Step Angle

Suppose, in a variable reluctance stepper motor the number of steps required is, n = 200. Then, the step angles $\Delta\theta$ can be calculated as

$$\Delta\theta = \frac{360°}{n} = 1.8° \qquad (3.2)$$

Example 3.3 Sequence for Full-step Angle

As explained in the beginning of Section 3.1.1, the full-stepping requires that one phase is energized at a time. For the two-phase motor of Fig. 3.3, the stepping sequence is given in Table 3.3. The rotor positions are shown in Fig. 3.8.

Table 3.3 Full-stepping sequence for a two-phase motor

Phases \ Step	1	2	3	4	Rotation
	→	→	→	→	CW
P_1	0	–1	0	1	
P_2	1	0	–1	0	
	←	←	←	←	CCW

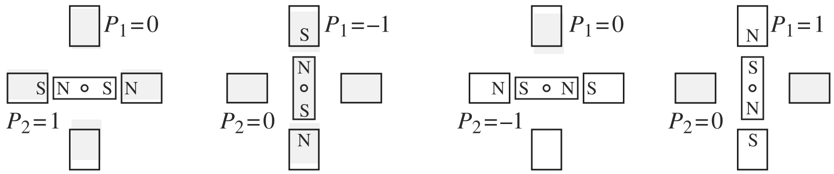


Fig. 3.8 Stepping sequence (one phase energized at a time)

3.1.2 dc Motors

People in their daily lives experience electric motors, whether they are dc (often battery operated) motors to run a child's toy or ac (mains operated) motors to turn the blenders in the kitchen, etc. The first commercial electrically driven industrial robot was introduced in 1974 by the Swedish giant corporation ASEA. Traditionally, roboticists have employed electrically driven dc (direct-current) motors for robots because, not only are powerful versions available, but they are also easily controllable with relatively simple electronics. Although direct current is needed, batteries are rarely used (for nonmobile robots) but instead the ac supply is *rectified* into a suitable dc equivalent. The operation of any electric motor is based upon the principle which states that a conductor will experience a force if an electric current in that conductor flows at right angles to a magnetic field. Therefore, to construct a motor, two basic components are required. One to produce the magnetic field usually termed the *stator*, and another to act as the conductor usually termed the *armature* or the *rotor*. The principle is shown in Fig. 3.9(a) for one element of a dc motor, whereas a two-pole dc motor is shown in Fig. 3.9(b). The magnetic field may be created either by *field coils* wound on the stator or by permanent magnets. The field coils, if used, would be provided with an electric current to create magnetic poles on the stator. In Fig. 3.9(a), the current is supplied to a conductor via the *brushes* and *commutators*. The current passing through the field produces a torque, or more accurately static torque on the conductor, which can be given by

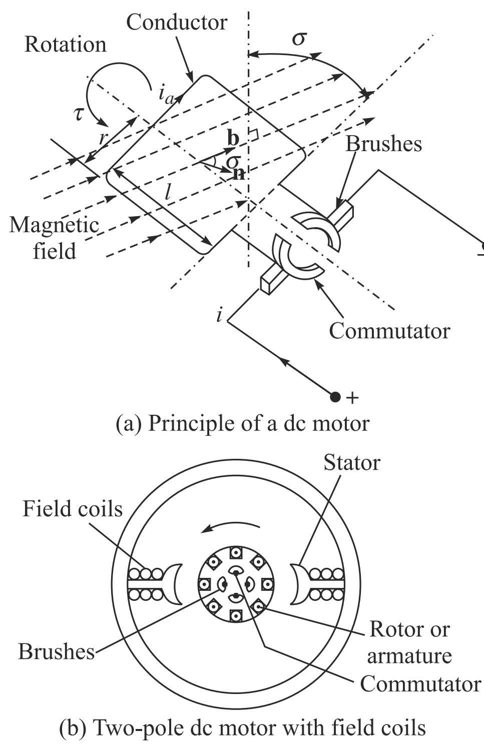


Fig. 3.9 dc motors

$$\tau = 2fr \sin \sigma \tag{3.3a}$$

where f is the magnetic force exerted on the conductor, r is the radius of the rotor, and σ is the angle between the flux density vector of stator magnetic field $\mathbf{b}$ and the unit vector normal to the plane of the coil $\mathbf{n}$, as shown in Fig. 3.9(a). The angle σ is known as *torque angle*. In view of Lorentz's law, the imbalanced magnetic force f that is generated on the conductor, normal to the magnetic plane is given by

$$f = Bi_a l \tag{3.3b}$$

in which B is the flux density of the original field, i_a is the current through the conductor, i.e., coil rotor or armature formed by the conductors, and l is the length of the conductor. Substituting Eq. (3.3b) into Eq. (3.3a), the torque τ generated in the rotor is given by

$$\tau = Ai_a b \sin \sigma \tag{3.3c}$$

In Eq. (3.3c), $A \equiv 2lr$, which is nothing but the face area of the planar rotor, as shown in Fig. 3.9(a). Moreover, unit vectors **b** and **n** represent the field direction and normal to the plane of the rotor.

According to Eq. (3.3c), τ is maximum when $\sigma = 90°$. Moreover, the higher the voltage supplied to the stator coils of the motor, the faster the motor turns, providing a very simple method of speed control. Similarly, varying the current to the armature controls the torque. Reversing the polarity of the voltage causes the motor to turn in the opposite direction. Some larger robots utilize field control dc motors, i.e., motors in which the torque is controlled by manipulating the current to the field coils. These motors allow high power output at high speeds and can give a good power-to-weight ratio. A typical specification of a dc motor is given in Table 3.4, whereas its speed-torque characteristic curve is shown in Fig. 3.10 for different voltage values.

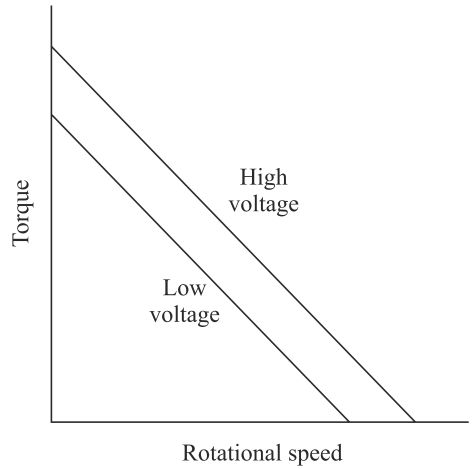


Fig. 3.10 Speed-torque speed characteristics of a dc motor

Table 3.4 Specifications of a dc motor [Courtesy: http://uk.rs-online.com]

Brand	Parvalux
Manufacturer part no.	PM2 160W511109
Type	Industrial dc Electric Motors
Shaft size (S, M, L)	M
Speed (rpm)	4000 rpm
Power rating (W)	160 W
Voltage rating (Vdc)	50 V(dc)
Input current	3.8 A
Height (mm)	78 mm
Width (mm)	140 mm
Length (mm)	165 mm

Note that the shape of steady-state speed-torque curves will be modified depending on how the windings of the rotor and the stator are connected. There are three typical ways to do that, namely, shunt-wound motor, series-wound motor, and compound-wound motor. Figure 3.11 shows the above three windings.

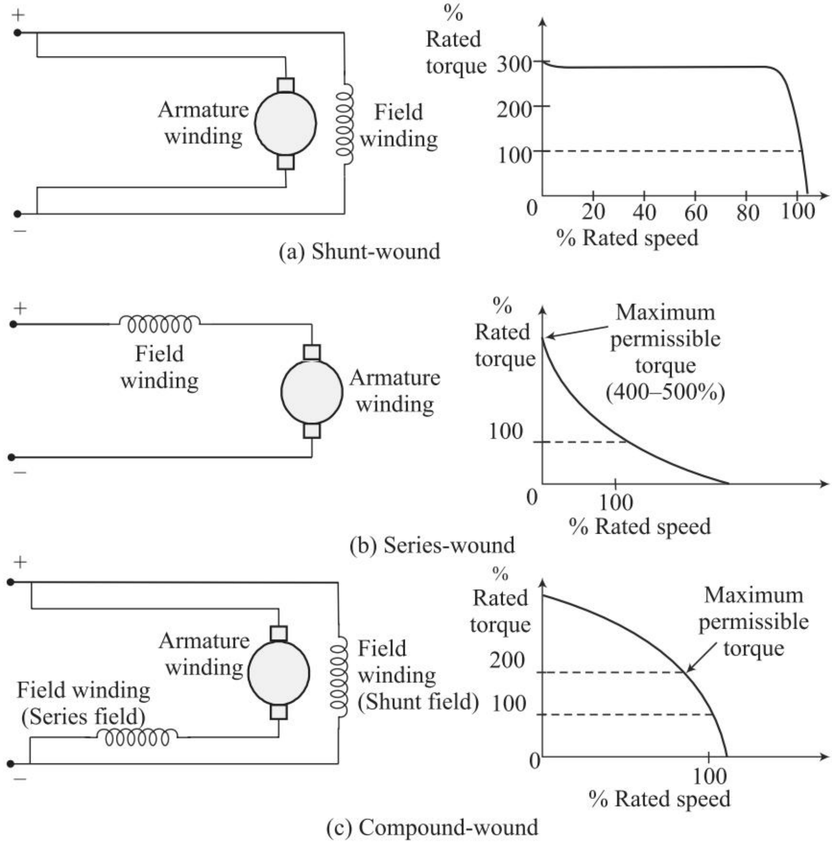


Fig. 3.11 Types of windings of dc motors

In a shunt-wound motor, the armature windings and field windings are connected in parallel, as shown in Fig. 3.11(a). At steady state, the back electromotive force (e.m.f.) depends directly on the supply voltage. Since the back e.m.f. is proportional to the speed, it follows that the speed controllability is good with the shunt-wound configuration. In series-wound motors, as the name suggests, they are connected in series. This is shown in Fig. 3.11(b). The relation between the back e.m.f. and supply voltage is coupled through both the armature windings and the field windings. Hence, its speed controllability is relatively poor. But in this case, a relatively large current flows through both windings at low speeds of the motor, giving higher starting torque. Also, the operation is approximately at constant power. In the compound-wound motor, a part of the field windings is connected with the armature windings in series and the other part is connected in parallel, as shown in Fig. 3.11(c). This kind of motors provides a compromise performance between the extremes of speed controllability and higher starting torque characteristics, as provided by the shunt-wound and series-wound motors, respectively. Table 3.5 summarizes the properties derived from the above three types of windings. For an industrial robot, in general, it is said that the current excited field control methods involve too slow a response time and incur losses that make permanent-magnet fields and armature control more attractive, which are explained next.

Table 3.5 Steady-state characteristics of a dc motor [Courtesy: de Silva, 2004]

Serial No.	Windings in a dc motor	Field-coil resistance	Speed controllability	Starting torque
1	Shunt-wound	High	Good	Average
2	Series-wound	Low	Poor	High
3	Compound-wound	Parallel high, series low	Average	Average

1. Permanent-Magnet (PM) dc Motors The permanent-magnet dc motor, which is also referred as *torque motor*, can provide high torque. Here, no field coils are used and the field is produced by the permanent magnets themselves. These magnets should have high-flux density per unit yielding a high torque/mass ratio. Typical materials with desired characteristic of such dc motors are rare-earth materials such as samarium cobalt, etc. Some PM motors do have coils wound on the magnet poles but these are simply to recharge the magnets if their strength fails. Due to the field flux being a constant, the torque of these motors is directly proportional to the armature current. Some other advantages are: excitation power supplies for the field coils are not required, reliability is improved as there are no field coils to fail, and no power loss from field coils means efficiency and cooling are improved. However, these types of motors are more expensive. Two types of PM configurations are shown in Fig. 3.12. They are cylindrical and disk types.

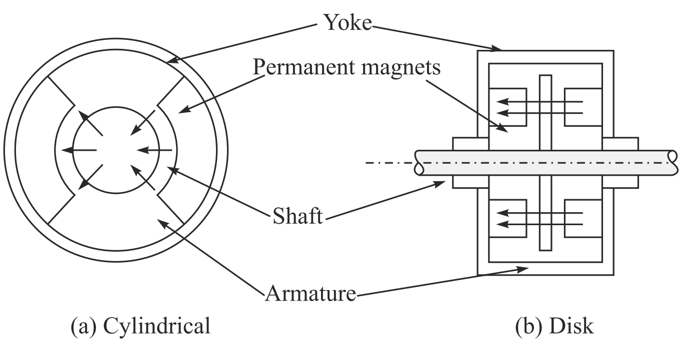


Fig. 3.12 Two types of permanent-magnet dc motor configurations

The cylindrical motor operates in a similar manner to that already described for other dc motors except that there are no field coils, whereas the disk motor has a large diameter, short-length armature of nonmagnetic material. It is the cylindrical motor that is more commonly used in industrial robots.

2. Brushless Permanent-Magnet dc Motors A problem with dc motors is that they require a commutator and brushes in order to periodically reverse the current through each armature coil. The brushes make sliding contacts with the commutators and as a consequence sparks jump between the two and they suffer wear. Brushes thus have to be periodically changed and the commutator resurfaced. To avoid such problems, brushless motors have been designed. Essentially, they consist of a sequence of stator coiled and a permanent magnet rotor. A current-carrying conductor in a magnetic field experiences a force; likewise, as a consequence of Newton's third law of motion, the magnet will also experience an opposite and equal force. With the conventional dc motor, the magnet is fixed and the current-carrying conductors made to move. With the brushless permanent-magnet dc motor the reverse is the case, the current-carrying conductors are fixed and the magnetic field moves. The rotor is a ferrite or ceramic permanent magnet. In concept, brushless dc motors are somewhat similar to permanent-magnet stepper motors explained in Section 3.1.1 and to some type of ac motors, as in Section 3.1.3. Figure 3.13 shows the basic form of such a motor. The current to the stator coils is electronically switched by transistors in sequence round the coils, the switching being controlled by the position of the rotor so that there are always forces acting on the magnet causing it to rotate in the same direction.

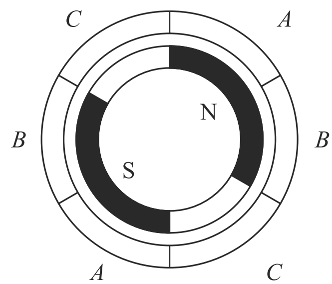


Fig. 3.13 Brushless permanent-magnet dc motor

The *brushless* motors have many advantages over conventional dc motors. For example,

- They have better heat dissipation, heat being more easily lost from the stator than the rotor.
- There is reduced rotor inertia. Hence, they weigh less and low mass for a specified torque rating.
- The motors in themselves are less expensive.
- They are more durable and have longer life.
- Low maintenance.
- Lower mechanical loading.
- Improved safety.
- Quieter operation.
- They are of smaller dimensions of comparable power.
- The absence of brushes reduces maintenance costs due to brush and commutator wear, and also allows electric robots to be used in hazardous areas with flammable atmospheres such as are found in spray-painting applications.

One disadvantage is that the control systems for brushless motors are relatively expensive due to sensing and switching.

3. dc Servomotors and their Drivers Servomotors are motors with motion feedback control, which are able to follow a specified motion trajectory. In a dc servomotor, both angular position and speed might be measured using, say, shaft encoders, tachometers, resolvers, potentiometers, etc., and compared with the desired position and speed. The error signal which is the difference between the desired minus actual responses is conditioned and compensated using analog circuitry or is processed by a digital hardware processor or a computer, and supplied to drive the servomotor toward the desired response. Motion control implies indirect torque control of the motor that causes the motion. In some applications like grinding, etc., where torque itself is a primary output, direct control of motor torque would be desirable. This can be accomplished using feedback of the armature current or the field current because those currents determine the motor torque. A typical layout of a dc servomotor is given in Fig. 3.14.

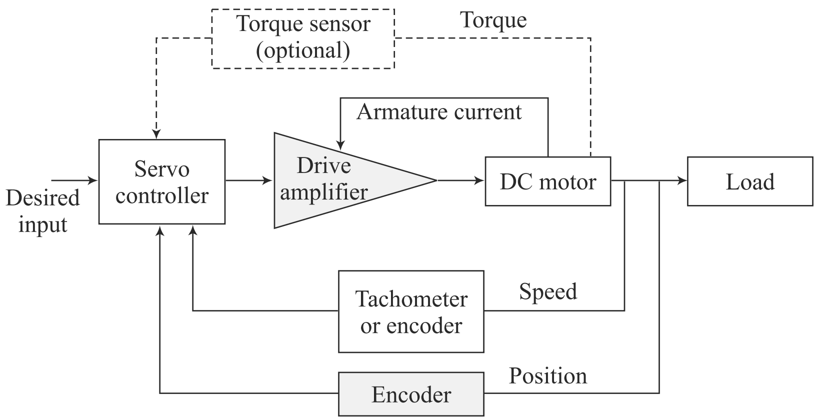


Fig. 3.14 A dc servomotor [Courtesy: de Silva, 2004]

Note that the control of a dc motor is achieved by controlling either the stator field flux or the armature flux. If the armature and field windings are connected through the same circuit, i.e., one of the winding types shown in Fig. 3.11, both techniques are implemented simultaneously. Two methods of control are armature control and field control. In armature control, the field current in the stator circuit is kept constant and the input voltage to the rotor is varied in order to achieve a desired performance. In the field control, on the other hand, the armature voltage is kept constant and input voltage to the field circuit is varied. These winding currents are generated using a *motor driver*. It is a hardware unit that generates necessary current to energize the windings of the motor. By controlling the current generated by the driver, the motor torque can be controlled. By receiving feedback from a motion sensor (encoder tachometer, etc.), the angular position and the speed of the motor can be controlled.

The drive unit of a dc servomotor primarily consists of a driver amplifier (commercially available amplifiers are linear amplifier or pulse-width-modulation, i.e., PWM, amplifier), with additional circuitry and a dc power supply. The driver is commanded by a control input provided by a host computer through an interface (input/output) card. A suitable arrangement is shown in Fig. 3.15. Also, the driver parameters like amplifier gains are

Impedance vs. Inertia Matching

In electrical systems, impedance matching is to equate the input impedance of a load (or output impedance) with the input impedance to maximize the power transfer. Similarly, in a motor-gearbox load configuration, the inertia (or mechanical impedance) of the load seen by the motor should be the same as its inertia in order to obtain maximum power transfer or acceleration.

software programmable and can be set by the host computer. The control computer receives a feedback signal of the motor motion, through the interface board, and generates a control signal, using a suitable control law explained in Chapter 10. The signal is then provided to the drive amplifier, again through the interface board. An interface board or Data Acquisition (DAQ) card is a hardware module with Digital-to-Analog (DAC) and Analog-to-Digital (ADC) capabilities built-in. These are generally parts of a robot's control system, as mentioned in Section 2.1.3.

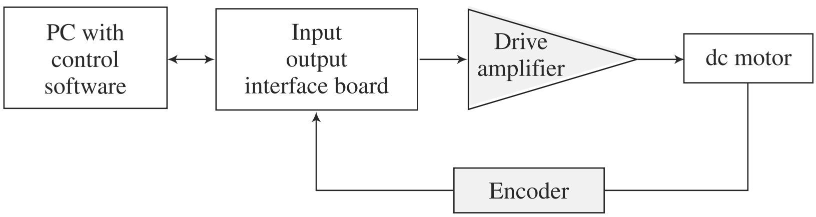


Fig. 3.15 Controller of a dc servomotor

The final control of a dc motor is accomplished by controlling the supply voltage to either the armature circuit or the field circuit. A dissipative method of achieving this involves using a variable resistor in series with the supply source to the circuit. This method has disadvantages of high heat generation, etc. Instead, the voltage to a dc motor is controlled by using a solid-state switch known as a *thyrister* to vary the off time of fixed voltage level, while keeping the period of pulse signal constant. Specifically, the duty cycle of a pulse signal is varied, as demonstrated in Fig. 3.16. This is called *pulse-width-modulation* or PWM. A commercial dc servomotor along with its controller unit is shown in Fig. 3.17.

Duty Cycle

It is given in percentage as

$$\frac{\text{On period}}{\text{Pulse period}} \times 100\%$$

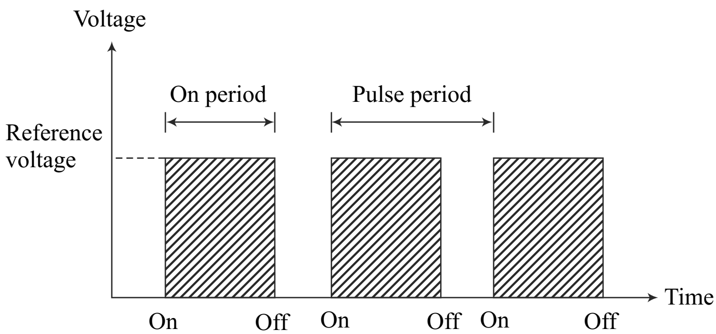


Fig. 3.16 Duty cycle of a PWM signal

(a) A dc servomotor

(b) Maxon's driver for its servomotor

[Courtesy: http://www.maxonmotor.in/maxon/view/content/ec-max-motors]

Fig. 3.17 A dc servomotor and its driver controller

Example 3.4 Stable and Unstable Operating Points of a dc Motor

Figure 3.18 shows the steady-state characteristic curve of a dc motor for a given load to be driven at constant power under steady-state operating condition using a separately wound dc motor with constant supply voltages to the field and armature windings. The constant power curve for the load is a hyperbola because $\tau\omega_m$ = constant. The two curves intersect at *A* and *B*. At *A*, if there is a slight decrease in the speed of operation, the motor (magnetic) torque will increase to τ_m^A and the load torque demand will increase to τ_l^A. But since $\tau_m^A > \tau_l^A$, the system will accelerate back to the point *A*. This shows that the point *A* is stable. On the other hand, at *B*, if speed drops slightly, the magnetic torque of the motor will increase to τ_m^B and at the same time, the load torque demand will also increase to τ_l^B, where $\tau_m^B < \tau_l^B$. As a result, the system will decelerate further leading to stalling the motor. Hence, the point *B* is unstable.

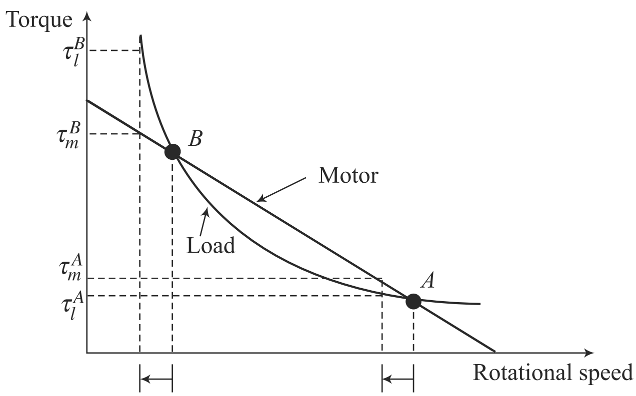


Fig. 3.18 Steady-state characteristic curve of a dc motor

3.1.3 ac Motors

Until recently, ac (alternating current) motors have not been considered suitable for robots because of the problems involved in controlling their speeds. In its simplest form, an ac motor consists of external electromagnets around a central rotor, but without any form of mechanical switching mechanism for the electromagnets. However, because alternating current (such as the mains electricity supply) is constantly changing polarity (first flowing one way, then the opposite way, several times a second, e.g., 50 in India, and 60 in the USA), it is possible to connect the ac supply directly to the electromagnets. The alternating reversal of the direction of current through the coils will perform the same task of polarity changing in ac motors. The magnetic field of the coils will appear to *rotate* (almost as if the coils themselves were being mechanically rotated).

A typical specification of an ac motor is given in Table 3.6. Moreover, speed-torque characteristics of ac motors are shown in Fig. 3.19. Typical advantages of an ac motor over its dc counterpart are listed below:

- Cheaper.
- Convenient power supply.
- No commutator and brush mechanism. Hence, virtually no electric spark generation or arcing (safe in hazardous environment like spray painting and others)
- Low power dissipation, and low rotor inertia.
- High reliability, robustness, easy maintenance, and long life.

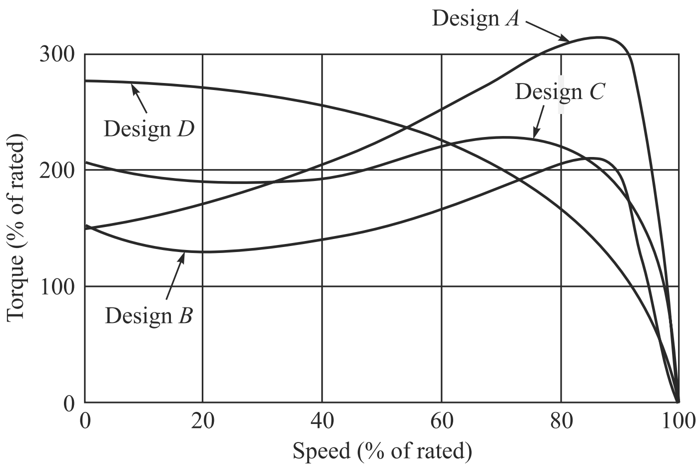
Some of the disadvantages are the following:

- Low starting torque.
- Need auxiliary devices to start the motor.
- Difficulty of variable-speed control or servo control unless modern solid-state and variable-frequency drives with field feedback compensation are used.

Table 3.6 Specifications of an ac motor [Courtesy: http://uk.rs-online.com]

Brand	ABB
Manufacturer part no.	1676687
Type	Industrial single and three-phase electric motors
Supply voltage	220 – 240 V_{ac} 50 Hz
Output power	180 W
Input current	0.783 A
Shaft diameter	14 mm
Shaft length	30 mm
Speed	1370 rpm
Rated torque	1.3 Nm
Torque starting	1.3 Nm
Height	150 mm
Length	213 mm
Width	120 mm

Due to the complexity in speed control of an ac motor, a speed-controlled dc drive generally works out cheaper than a speed-controlled ac drive, though the price difference is steadily dropping as a result of technological developments and the reduction in price of solid-state devices. A brief account of an ac servo motor is given at the end of this subsection.



[Courtesy: www.electricmotors.machinedesign.com/guiEdits/Content/bdeee11/bdeee11_7.aspx]

Fig. 3.19 Typical speed-torque characteristics for four different designs of an ac induction motor

Alternating current (ac) motors can be classified into two groups, single-phase and poly-phase, with each group being further subdivided into *induction* or *asynchronous* and *synchronous* motors. Single-phase motors tend to be used for low power requirements while poly-phase motors are used for higher powers. Induction motors tend to be cheaper than synchronous motors and are thus very widely used.

1. Single-phase Squirrel-cage Induction Motor It consists of a squirrel-cage rotor, this being copper or aluminum bars that fit into slots in end rings to form complete electrical circuits, as shown in Fig. 3.20. There are no external electrical connections to the rotor. The basic motor consists of this rotor with a stator having a set of windings. When an alternating current passes through the stator windings, an alternative magnetic field is produced, which appears like a rotating magnetic field. The rotating field in the stator intercepts the rotating windings, thereby generating an induced current due to mutual induction or transformer action (hence, the name induction motor). The resulting secondary magnetic flux interacts with the primary, rotating magnetic flux, thereby producing a torque in the direction of rotation of the stator field. This torque drives the rotor. As the rotor speed increases, initially the motor torque also increases because of secondary interactions between the stator circuit and the rotor circuit even though the relative speed of the rotating field with respect to the rotor decreases, which reduces the rate of change of flux linkage and,

> **dc motor or ac motor?**
>
> The dc motors are easy to control compared to the ac motors. Hence, the former is preferred in robotic applications.

hence, the direct transformer action. Note that the relative speed is termed as slip rate given by

$$S = \frac{\omega_f - \omega_m}{\omega_f} \tag{3.4}$$

where ω_f and ω_m are the speed of the field and motor's rotor, respectively. Quite soon, the maximum torque will be reached. Further increase in the rotor speed (i.e., a decrease in slip rate) sharply decreases the motor torque, until at synchronous speed (zero slip rate), the motor torques becomes zero.

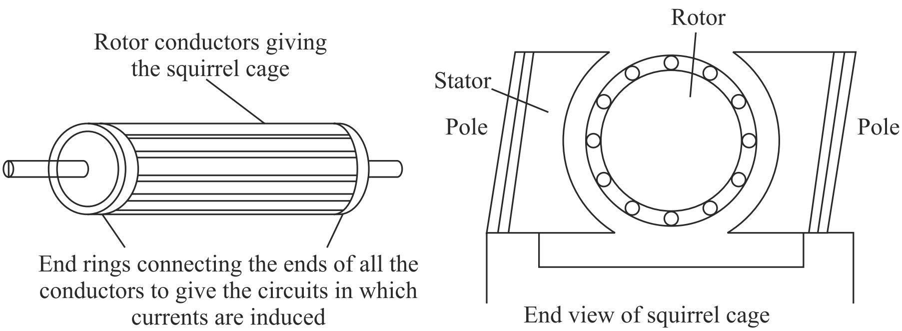


Fig. 3.20 Single-phase induction motor

For a single-phase supply, when the rotor is stationary initially, the forces on the current-carrying conductors or the rotor in the magnetic field of the stator are such as to result in no net torque. Hence, the motor is not self-starting. A number of methods are used to make the motor self-starting and give this initial impetus to start it. For example, to provide the starting torque, most single-phase motors have a main and auxiliary winding. The auxiliary winding current from the main winding is phase-shifted. Connecting a capacitor in series with the auxiliary winding causes the motor to start rotating. The rotor rotates at a speed determined by the frequency of the alternating current applied to the stator. For a constant frequency supply to a two-pole single-phase motor, the magnetic field will alternate at this frequency. This speed of rotation of the magnetic field is termed *synchronous speed.* The rotor will never quite match this frequency of rotation, typically differing from it by about 1 to 3%. For a 50 Hz supply, the speed of rotation of the rotor, i.e., ω_m, will be almost 50 revolutions per second.

Not Self-Starting of a Single-Phase Motor?

The stator winding of a single-phase motor can only produce the field (flux), which is only alternating but not rotating. Hence, it cannot produce any rotation.

2. Three-phase Induction Motor As shown in Fig. 3.21(a), it is similar to the single-phase induction motor but has a stator with three windings located 120° apart, each winding being connected to one of the three lines of the supply. Because the three phases reach their maximum currents at different times, the magnetic field can be considered to rotate round the stator poles, completing one rotation in one full cycle of the current. The rotation of the field is much smoother than with the single-phase motor. The three-phase motor has a great advantage over the single-phase motor in being self-starting. The direction of rotation is reversed by interchanging any two of the line connections, thus changing the direction of rotation of the magnetic field.

3. Synchronous Motor A synchronous motor has a stator similar to those described above for induction motors, but a rotor is a permanent magnet as shown in Fig. 3.21(b). The magnetic field produced by the stator rotates and so the magnet rotates with it. With one pair of poles per phase of the supply, the magnetic field rotates through 360° in one cycle of the supply and so the frequency of rotation with this arrangement is the same as the frequency of the supply. Synchronous motors are used when a precise speed is required. They are not self-starting and some system has to be employed to start them.

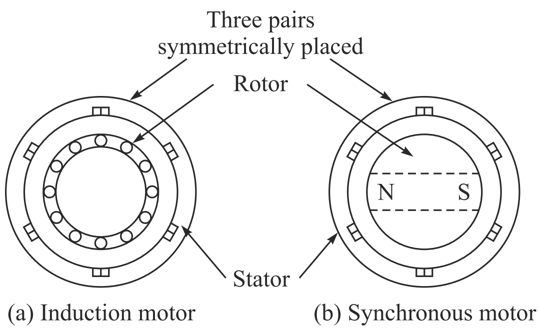


Fig. 3.21 An ac three-phase motor

4. ac Servomotor Generally, a modern servomotor refers to an ac permanent-magnet synchronous servomotor. It consists of a stator winding plus rotor with feedback units like encoders, resolvers, etc. These motors have typical advantages of ac motors as mentioned in the beginning of Section 3.1.3. Speed control of ac motors is based on the provision of a variable frequency supply, since the speed of such motors is determined by the frequency of the supply. In principle, the higher the frequency of the alternating current applied to the motor, the faster it will rotate. Providing varying frequency supplies to a number of axis drives simultaneously has been, until recently, largely impractical. In some special cases, e.g., wound-rotor ac induction motors, speed can be controlled by accessing the rotor circuit where different values of resistance can be inserted in the rotor circuit. Electromagnetic is used to provide regenerative braking to cut down the deceleration times, and minimize axis overrun. Many industrial robots, e.g., KUKA KR-5 of Fig. 2.1, use ac servomotors today. A typical ac servomotor is shown in Fig. 3.22, whereas its servo controllers look like those shown in Fig. 2.9(c) for the KUKA robot. Typical specification of an ac servomotor used is shown in Table 3.7.

[Courtesy: http://motorsforautomation portal.info/kuka-robot-servo-motor/]

Fig. 3.22 A typical ac servomotor

Table 3.7 Specifications of an ac servo motor [Courtesy: http://uk.rs-online.com]

Brand	Siemens
Manufacturer part no.	1FK7042-5AF21-1PA3
Type	ac synchronous servomotor
Supply voltage	230 V 50 Hz
Power rating	820 W
Maximum output torque	2.6 Nm
Output speed	3000 rpm
Shaft diameter	19 mm
Stall torque	3 Nm
Height	96 mm
Length	162 mm
Width	90 mm
Encoder system for motor	14-bit resolver (with DRIVE-CLiQ-interface)

3.1.4 Linear Actuators

Linear actuators like solenoids are used widely in robotic and other automation applications for on-off of the gripper and other devices. Electrically powered stepper and dc/ac linear actuators can also be used in motion generation of Cartesian robots, etc.

A solenoid shown in Fig. 3.23 has a coil and a soft-iron core. When the coil is activated by a dc signal, the soft core becomes magnetized. This electromagnet can serve as on-off (push-pull) actuator. Solenoids are rugged and inexpensive devices. Common applications of solenoids are valve actuators (as explained in Section 3.2 and 3.3 for fluid-power actuators), mechanical switches, relays, etc. Most commonly generated linear

Backlash

Backlash is the term used to describe the unwanted play in transmission components from their intended position, when put under load conditions, due to wear or clearances between surfaces.

motions are with the help of an electrically powered rotary motor, as explained in sections 3.1.1–3.1.3, coupled with transmission mechanisms like nut and ball-screw, cam-follower, rack-and-pinion, etc. These devices inherently have problems of friction and backlash. Additionally, they add inertia and flexibility to the driven load, thereby generating undesirable resonances and motion errors. Proper inertia matching is also essential. In order to avoid the above difficulties of using a transmission system, direct rectilinear electromechanical actuators are desirable. They can be based on any of the principles mentioned for the rotary actuators, i.e., stepper or dc or ac motors. In these actuators, flat stators and rectilinearly moving elements (in place of rotors) are employed. These types of actuators are also referred to as electric cylinders in line with hydraulic or pneumatic cylinders that are explained next.

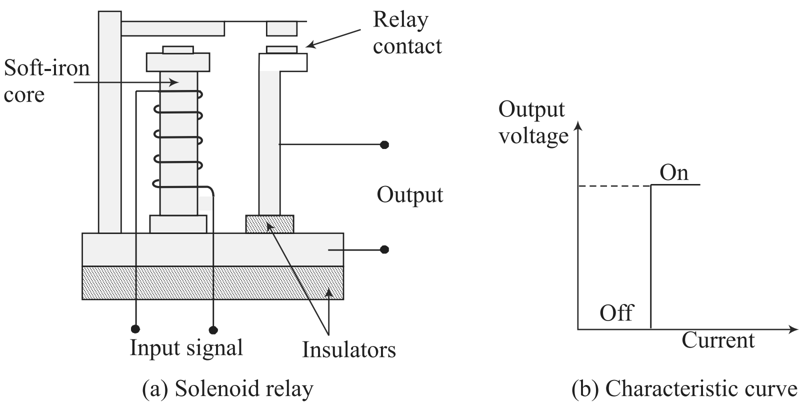


Fig. 3.23 A solenoid and its characteristic curve

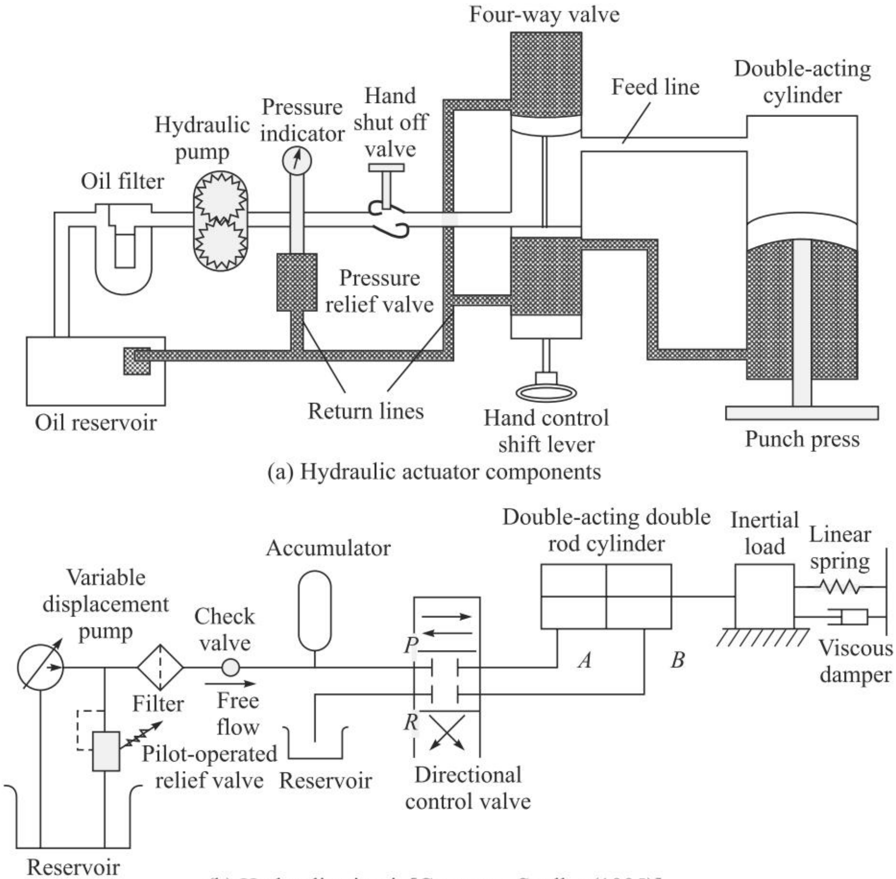
3.2 HYDRAULIC ACTUATORS

> **What is 1 Bar?**
>
> It is a unit of pressure.
> One Bar = one atm = 14.7 Psi = 101356.5 N/m^2 (Pa) or approximately 0.1 MPa.

Hydraulic actuators are one of the two types of fluid power devices for industrial robots. The other type is pneumatic, described in the next section. Hydraulic actuators utilize high-pressure fluid such as oil to transmit forces to the point of application desired. A hydraulic actuator is very similar in appearance to that of pneumatically driven one. Different components are shown in Fig. 3.24, whereas typical specifications of a cylinder are given in Table 3.8. Hydraulic actuators designed to operate at much higher pressures (typically between 70 and 170 bar). They are suitable for high power applications. Advantages and disadvantages of a hydraulic actuator are as follows:

Advantages

- High efficiency and high power-to-size ratio.
- Complete and accurate control over speed, position, and direction of actuators are possible.



(b) Hydraulic circuit [Courtesy: Stadler (1995)]

Fig. 3.24 A hydraulic actuator

- Few backlash problems occur due to the stiffness and incompressibility of the fluid, especially, when the actuator acts as the joint itself. Large forces can be applied directly at the required locations.
- They generally have a greater load-carrying capacity than electric and pneumatic robots.
- No mechanical linkage is required, i.e., a direct drive is obtained with mechanical simplicity.
- Self-lubricating (low wear) and non-corrosive.
- Due to the presence of an accumulator, which acts as a 'storage' device, the system can meet sudden demands in power.
- Hydraulic robots are more capable of withstanding shock loads than electric robots.

Disadvantages

- Leakages can occur to cause loss in performance, and general contamination of the work area. There is also higher fire risk.

- The power pack can be noisy, typically about 70 decibel (dBA) or louder if not protected by an acoustic muffler.
- Changes in temperature alter the viscosity of the hydraulic fluid. Thus, at low temperatures, fluid viscosity will increase, possibly, causing sluggish movement of the robot.
- For smaller robots, hydraulic power is usually not economically feasible as the cost of hydraulic components do not decrease in proportion to size.
- Servo control of hydraulic systems is complex and is not as widely understood as electric servo control.

How noisy is 70 decibels?

The level of 70 decibels is about the noise level of heavy traffic.

Hydraulic systems power the strongest and the stiffest robots and, hence, the bulk modulus of the oil is an extremely important attribute to be selected. A high bulk modulus implies a stiff, quickly responding system with a corresponding quick pressure build-up, while a low bulk modulus may result in a system that is too loose because of the high compressibility of the oil. Hydraulic systems or circuits have always four essential components: a reservoir to hold the fluid, pumps to move it, valves to control the flow, and an actuator to carry out the dictates of the fluid on some load. Rotary hydraulic actuators are also available in the market. A commercially available hydraulic cylinder is shown in Fig. 3.25(a), whereas its design using software is available in WR-Hydraulic (2013). A video of such cylinders is also available at WR-Hydraulic-video (2013).

Table 3.8 Specifications of a hydraulic cylinder [Courtesy: http://uk.rs-online.com]

Brand	Rexroth
Manufacturer part no.	CDL1M00/32/18/50/C1X/B1CHUMS
Type	Cylinder-rod ends bearing
Body	Carbon steel
Bore size	32 mm
Stroke	50 mm
Port size	G 1/4
Static proof pressure	240 bar
Working pressure	160 bar
Maximum force at 160 bar	12.8 kN
Maximum flow at 0.1 mm/s	4.8 l/min
Overall length	200 mm
Overall length (piston extended)	250 mm
Rod-end diameter	18 mm
Rod-end thread size	18 mm
Seal material	NBR/Pollyurethane
Recommended hydraulic medium	Mineral-oil-based fluids
Hydraulic fluid operating temperature range	–20 to +80o C

[Courtesy: www.meritindustriesltd.com]

(a) Hydraulic cylinder

[Courtesy: www.festo.com]

(b) Pneumatic cylinder

Fig. 3.25 Commercially available cylinders

3.3 PNEUMATIC ACTUATORS

Pneumatic actuators are the other type of fluid power devices for industrial robots. Pneumatic actuators utilize compressed air for the actuation of cylinders as shown in Fig. 3.25 (b), and are widely used for typical opening and closing motions of jaws in the gripper of a robot, as shown in Fig. 2.4 (a), or for the actuation of simple robot arms used in applications where continuous motion control is not of concern. A pneumatic actuator comprising of a pneumatic cylinder and other accessories are shown in Fig. 3.26, whereas the specifications of a cylinder are given in Table 3.9. Typical advantages and disadvantages of pneumatic actuators are as follows:

Advantages

- It is the cheapest form of all actuators. Components are readily available and compressed air is normally an already existing facility in factories.
- Compressed air can be stored and conveyed easily over long distances.
- Compressed air is clean, explosion-proof and insensitive to temperature fluctuations, thus, lending itself to many applications.
- They have few moving parts making them inherently reliable and reducing maintenance costs.
- Since pneumatic systems are common throughout industry, therefore, relevant personnel are often very familiar with the technology.
- Very quick in action and response time, thus, allowing fast work cycles.
- No mechanical transmission is usually required.
- Pneumatics can be intrinsically safe in explosive areas as no electrical control is required. Also in wet conditions there is no danger of electrocution.
- The systems are usually compact.
- Control is simple, e.g., mechanical stops are often used.
- Individual components can be easily interconnected.

Disadvantages

- Since air is compressible, precise control of speed and position is not easily obtainable unless more complex electromechanical devices are incorporated into the system. This means that only a limited sequence of operation at a fixed speed is often available.

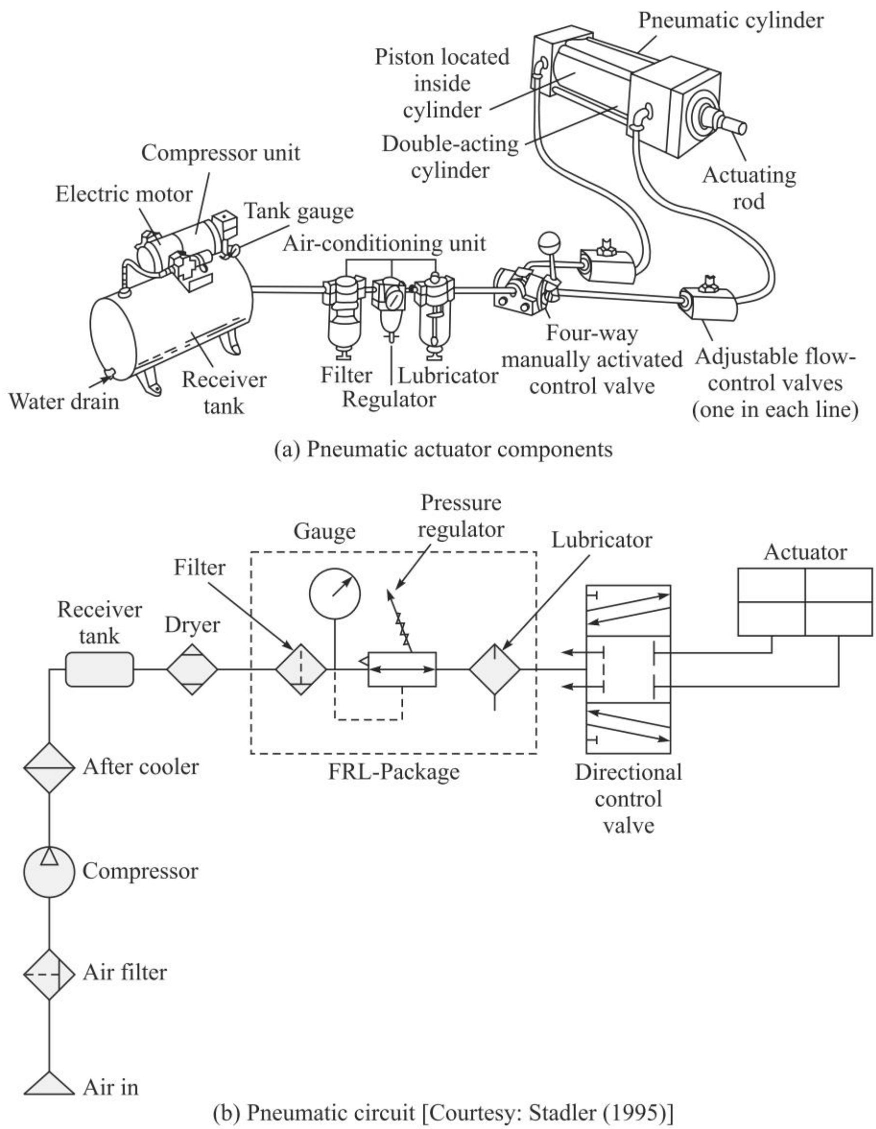


Fig. 3.26 A pneumatic actuator

- If mechanical stops are used, resetting the system can be slow.
- Pneumatics is not suitable for moving heavy loads under precise control due to the compressibility of air. This compressibility necessitates the application of more force than would normally be necessary to ensure that the actuator is firmly in position against its stop under load conditions.
- If moisture penetrates the units and ferrous metals have been used then damage to individual components can result.

The compressibility of the air in a pneumatic actuator does not allow for sophisticated control, but the same can be used to advantage, preventing damage due to overload and providing compliance that may be required in many practical applications.

Table 3.9 Specifications of a pneumatic cylinder for a gripper [Courtesy: http://uk.rs-online.com]

Brand	SMC
Manufacturer part no.	MHC2-25D
Type	Gripper (double acting)
Bore	25 mm
Operating angle	30° (open) to –10° (Closed)
Pressure range	1 to 6 Bar
Connection port size	M5 m
Temperature range	0 to 60°C
Media	Non lubricated air
Holding moment	1.4 Nm at 5 bar
Repeatability at closing position	±0.1 mm
Maximum operating frequency	80 cpm (cycles per minute)

3.4 SELECTION OF MOTORS

For any application of robots, one must decide which of available actuators is most suitable. Positioning accuracy, reliability, speed of operation, cost, and other factors must be considered. Electric motors are inherently clean and capable of high precision if operated properly. In contrast, pneumatic systems are not capable of high precision for continuous-path operation and hydraulic actuators require the use of oil under pressure. Hydraulic system can generate greater power in a compact volume than electric motors. Oil under pressure can be piped to simple actuators capable of extremely high torque and rapid operations. Also, the power required to control an electro-hydraulic valve is small. Essentially, the work is done in compressing the oil and delivering it to the robot-arm drives. All the power can be supplied by one powerful, efficient electric motor driving the hydraulic pump at the base of the robot or located at some distance away. Power is controlled in compact electro-hydraulic valves. However, high-precision electro-hydraulic valves are more expensive and less reliable than low-power electric amplifiers and controllers. On the other hand, electric motors must have individual controls capable of controlling their power. In large robots, this requires switching of 10 to 50 amperes at 20 to 100 volts. Current switching must be done rapidly; otherwise there is large power dissipation in the switching circuit that will cause excessive heat. Small electric motors use simple switching circuits and are easy to control with low-power circuits. Stepper motors are especially simple for open-loop operation. The single biggest advantage of a hydraulic system is their intrinsically safe operation.

As a rule of thumb, hydraulic actuators are preferred where rapid movement at high torques is required, at power ranges of approximately over 3.5 kW unless the slight possibility of an oil leakage cannot be tolerated. Electric motors are preferred at power levels under about 1.5 kW unless there is danger due to possible ignition of explosive materials. At ranges between 1-5 kW, the availability of a robot in a particular coordinate system with specific characteristics or at a lower cost may determine the decision. Reliability of all types of robots made by reputable manufacturers is sufficiently good that this is not a major determining factor.

Example 3.5 **Selection of a Motor**

Simple mathematical calculations are needed to determine the torque, velocity, and power characteristics of an actuator or a motor for different applications. Torque is defined in terms of force times distance or moment. A force, f, at distance, a, from the center of rotation has a moment or torque, τ, i.e., $\tau = fa$. In general terms, power, P, transmitted in a drive shaft is determined by the torque, τ, multiplied by the angular velocity, ω. Power P is expressed as, $P = \tau\, \omega$. For an example, a calculation can tell one what kilowatt or horsepower is required in a motor used to drive a 2-meter robot arm lifting a 25 kg mass at 10 rpm. If the mass of the arm is assumed zero then, $P = (25 \times 9.81 \times 2) \times (2\pi \times 10/60) = 0.513$ kW. The use of simple equations of this type is often sufficient to make a useful approximation of a needed value. More detailed calculations can take place using the equations of statics and dynamics that apply.

3.5 GRIPPERS

Grippers are end-effectors, as introduced in Section 2.1.1, which are used to grasp an object or a tool, e.g., a grinder, and hold it. Tasks required by the grippers are to hold workpieces and load/unload from/to a machine or conveyer. Grippers can be mechanical in nature using a combination of mechanisms driven by electric, hydraulic, or pneumatic powers, as explained in earlier sections. Grippers can be classified based on the principle of grasping mechanism. For example, grippers can hold with the help of suction cups, magnets, or by other means. A gripper is then accordingly referred to as *pneumatic gripper*, *magnetic gripper*, etc. Another way to classify a gripper is based on how it holds an object, i.e., based on grasping the object on its exterior (*external gripper*) or interior (*internal gripper*) surface.

3.5.1 Mechanical Grippers

As shown in Fig. 2.4(a), mechanical grippers have their jaw movements through pivoting or translational motion using a transmission element, e.g., linkages or gears, etc. This is illustrated in Fig. 3.27. The gripper can be of single or double type. While the former has only one gripping device at the robot's wrist, the latter type has two. The double grippers can be actuated independently and are especially useful in machine loading and unloading. As illustrated in Groover et al. (2012), suppose a particular job calls for a raw part to be loaded from a conveyor onto a machine and the finished part to be unloaded onto another conveyor. With a single gripper, the robot would have to unload the finished part before picking up the raw part. This would consume valuable time in the production cycle because the machine would remain idle during these handling motions. With a double gripper, the robot can pick up the part from the incoming conveyor with one of its gripping devices and have it ready to exchange for the finished part on the machine. When the machine cycle is completed, the robot can reach in for the finished part with the available grasping